

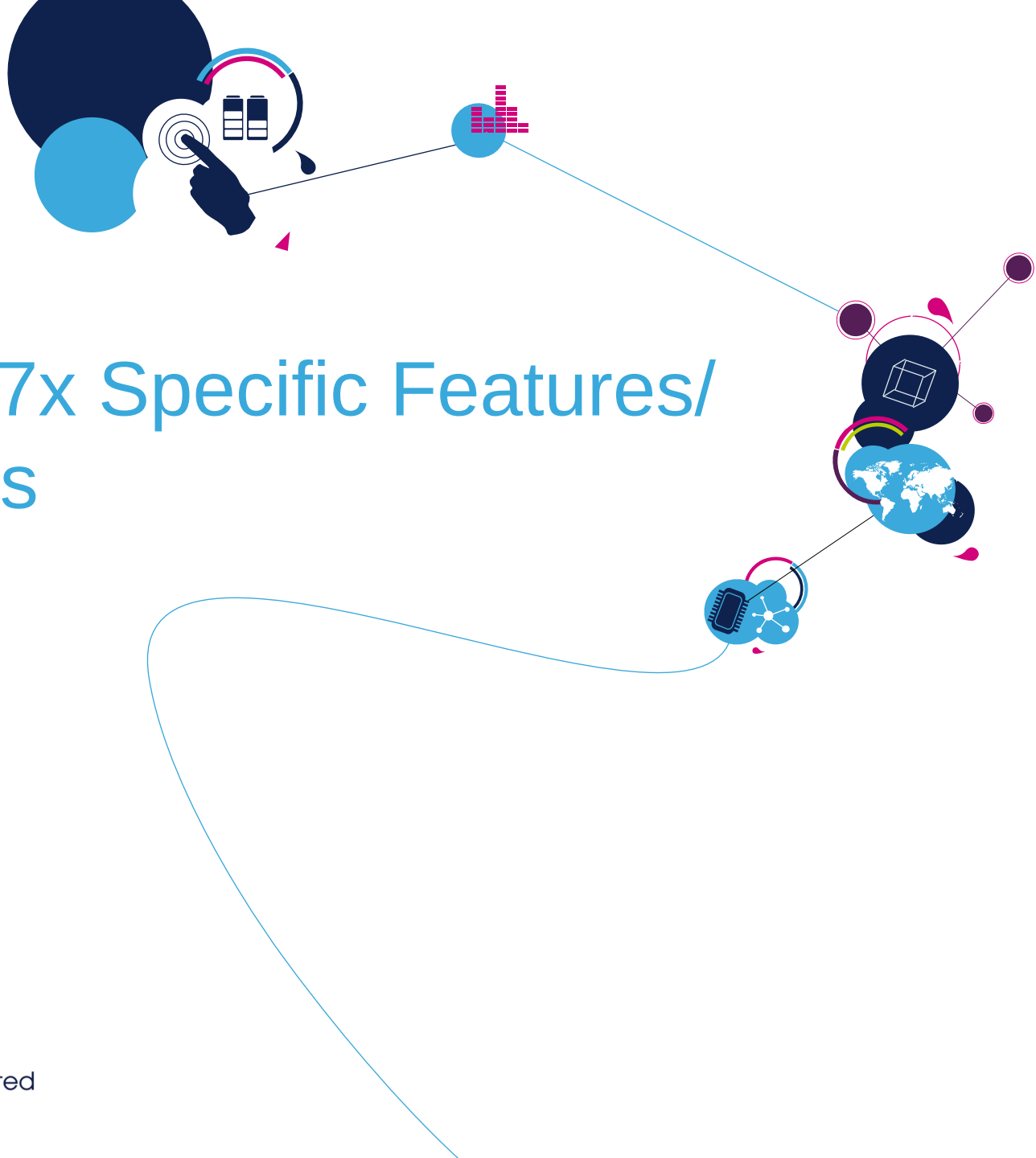


STM32F3 Technical Training

For reference only

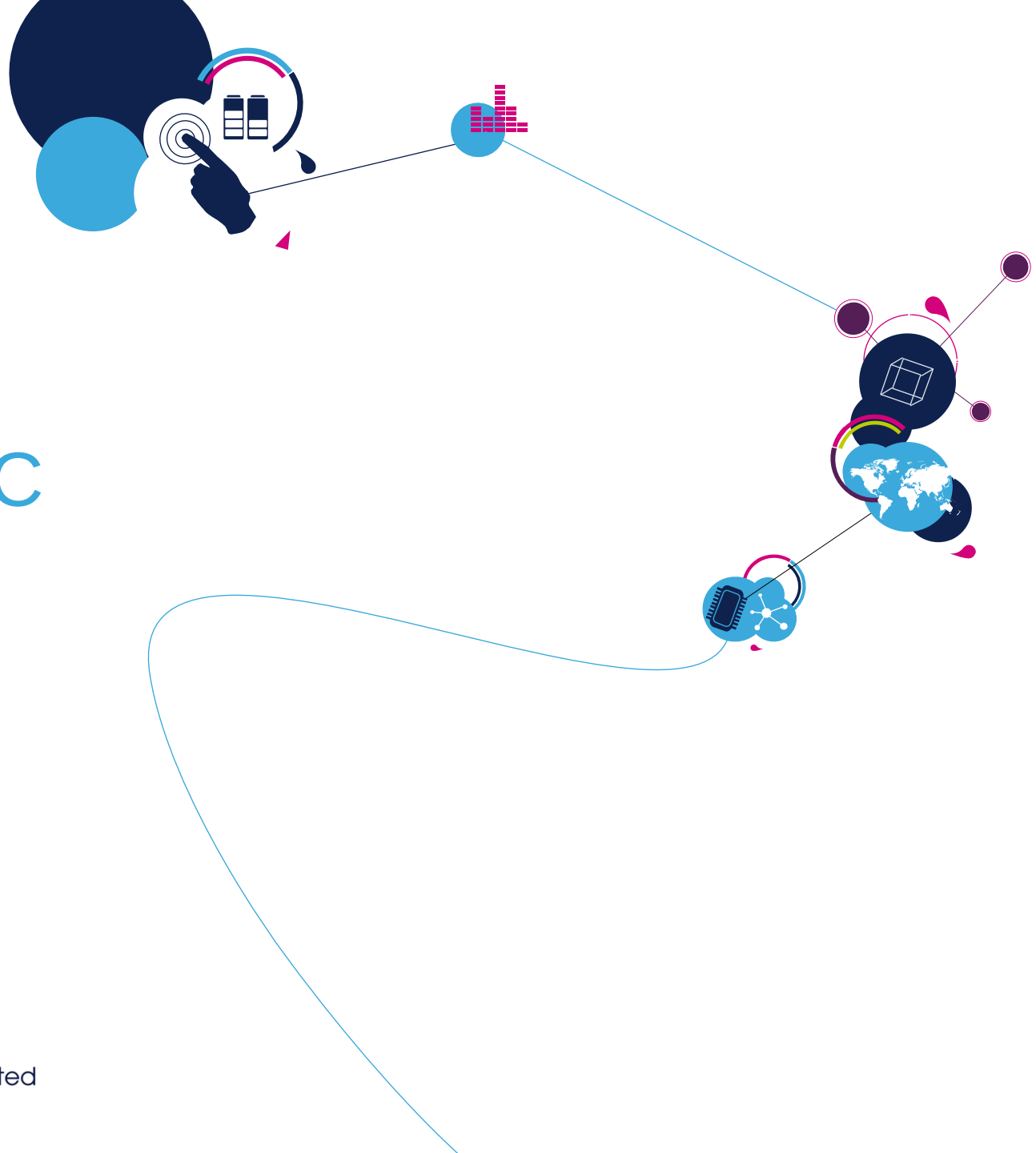
Refer to the latest documents for details





STM32F37x Specific Features/ peripherals

HDMI-CEC



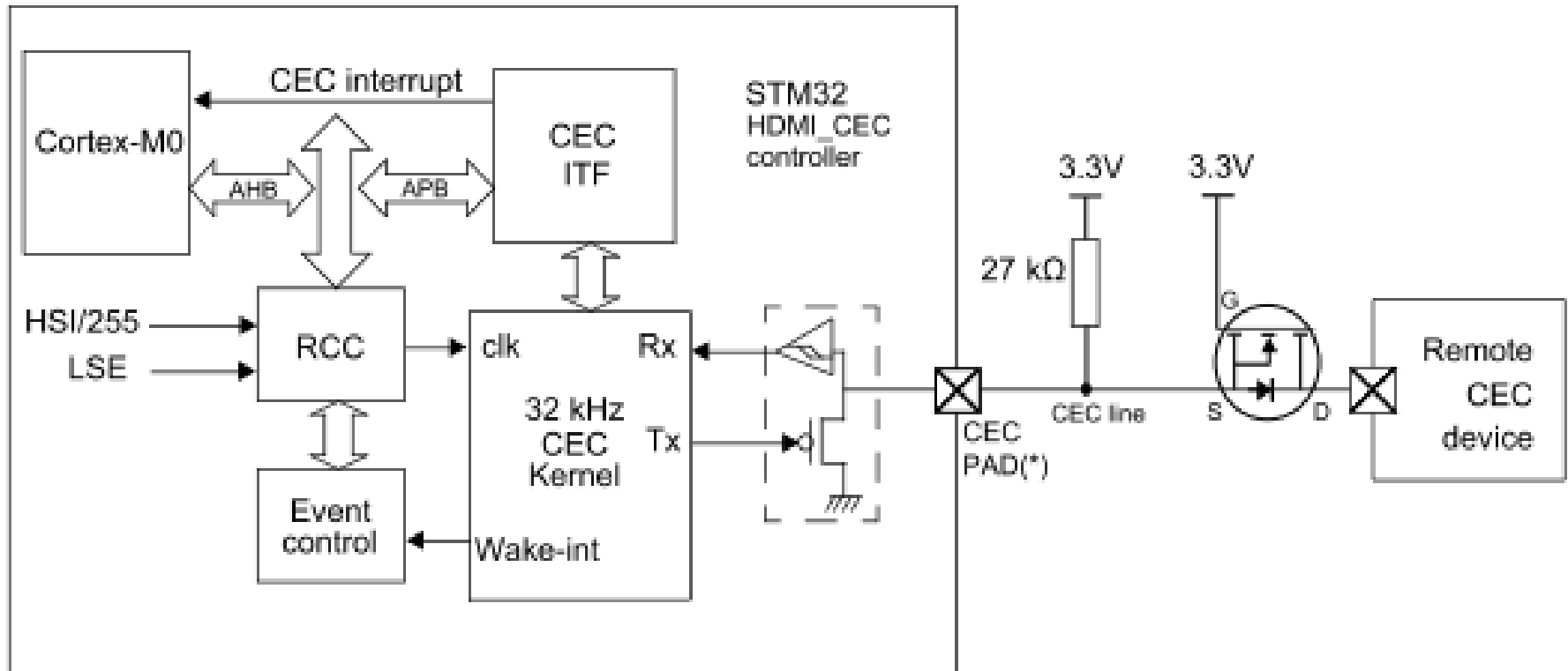
HDMI-CEC v2 Controller Features

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- Fully compatible with HDMI-CEC v1.3a standard
 - Electrical specifications
 - Messages (Frame formats, bits timings...)
 - Full Arbitration: Signal Free Time (SFT), Header Arbitration
- 32kHz kernel running from LSE or HSI/244 with wakeup from STOP
- Multiple logical addresses support + listen mode
- Configurable error handling with selectable extended timing tolerance
- Selectable signal free time (SFT) before transmission – HW or SW
- CEC line needs an external 27k Ω pull up and optional isolation

HDMI-CEC Controller block diagram

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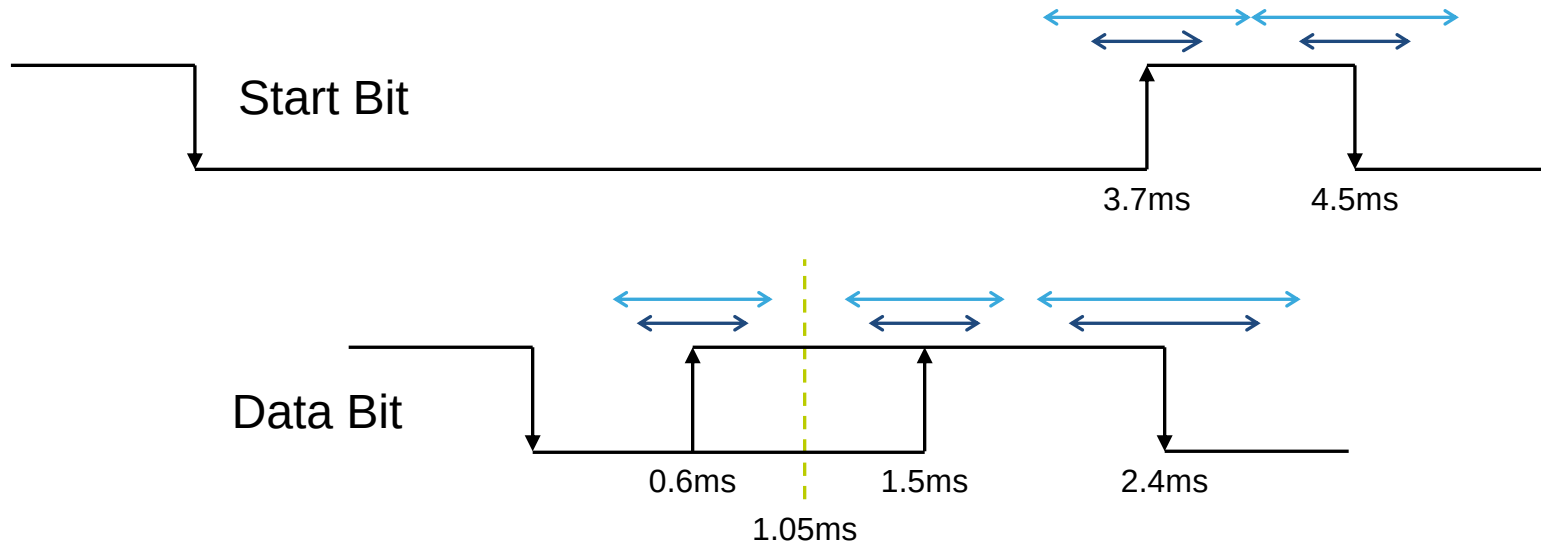
HDMI-CEC Interrupts 6

- An interrupt is triggered:
 - if a receive block transfer completes
 - if a transmit block transfer completes
 - in case of any receive or transmit error
 - in case of RX or TX buffer overrun or underrun
 - for transmission or reception end
 - in case of arbitration lost

Rx-Byte Received	RXBR
End of reception	RXEND
Rx-Overrun	RXOVR
RxBit Rising Error	BRE
Rx-Short Bit Period Error	SBPE
Rx-Long Bit Period Error	LBPE
Rx-Missing Acknowledge Error	RXACKE
Arbitration lost	ARBLST
Tx-Byte Request	TXBR
End of transmission	TXEND
Tx-Buffer Underrun	TXUDR
Tx-Error	TXERR
Tx-Missing Acknowledge Error	TXACKE

RX tolerance margins

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RxTol bit

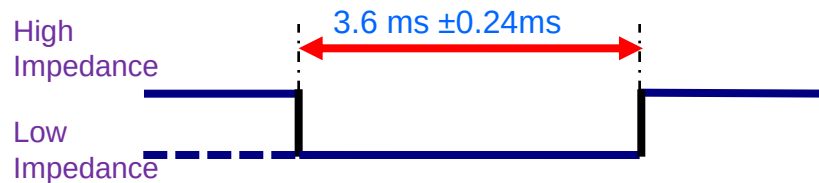
0b: Standard tolerance (in line with CEC specification)

Start bit: ± 200 rise & fall. Data bit: ± 200 rise, ± 350 fall

1b: Extended tolerance

Start bit: ± 400 rise & fall. Data bit: ± 300 rise, ± 500 fall

- CEC specifications says only:
 - It is the responsibility of all devices acting as followers to detect the existence of spurious pulses on the control signal line and notify all other devices (primarily the initiator) that a potential error has occurred.
 - An error is defined as a period between falling edges that is less than a minimum data bit period (i.e. too short to be a valid bit).
- Other timing errors are not considered in CEC specification – user define the action
- The error notification (error bit) is a low period on the CEC line of 1.4 to 1.6 times the nominal data bit period, that is, 3.6 ms nominally:

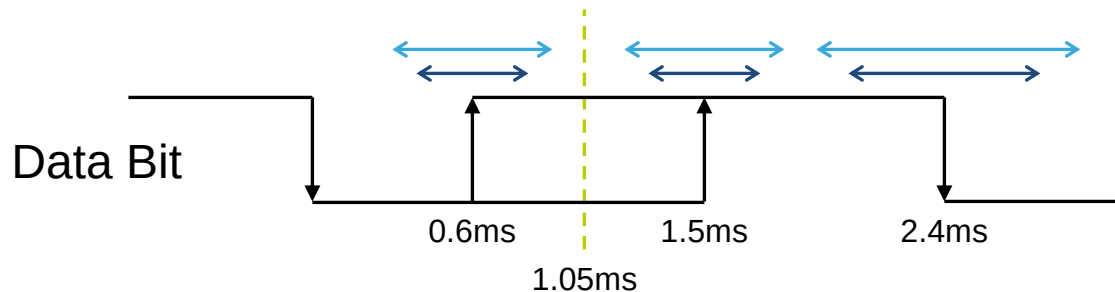


- A message is considered lost and therefore may be retransmitted under the following conditions:
 - a message is not acknowledged in a directly addressed message
 - a message is negatively acknowledged in a broadcast message
 - a low impedance is detected on the CEC line when not expected (line error)

- BRE: Bit Rising Error
 - BRE is set by HW at the time a **rising edge** is detected within a data bit **outside of** the Rx-**windows** configured by RxTol. Upon BRE detection CEC message reception is optionally aborted if BRESTOP=1 and **Error bit is optionally generated** on the CEC line if BREGEN=1.
- SBPE: Short Bit Period Error
 - SBPE is set by HW when a **falling edge** is detected ending the data bit **before** than **expected** by the RxTol margin. Upon SBPE detection an **Error bit is always generated** on the CEC line and reception aborted. CEC starts waiting for next start-bit once the CEC line is idle again.
- LBPE: Long Bit Period
 - LBPE is set by HW either when a **rising or falling** edge is detected **after the maximum RxTol margin**. Upon LBPE detection message **reception is always aborted** and an **Error bit optionally generated** on the CEC line if LBPEGEN=1.

Bit Timing Error detection

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Rising Edge	0.0	0.3	0.4	0.8	0.9	1.2	1.3	1.7	1.8
RxTol=0		BRE	'1'		BRE		'0'		BRE
RxTol=1		BRE	'1'			BRE	'0'		BRE

Falling Edge	0.0	0.3	0.4	0.8	0.9	1.2	1.3	1.7	1.8	1.9	2.05	2.75	2.9
RxTol=0												Ok	LBPE
RxTol=1												Ok	LBPE

Bit Timing Error configured action 1/2

- **BREGEN: Generate Error-Bit on Bit Rising Error**
 - 0: BRE detection **does not generate** Error-Bit on the CEC line.
 - 1: BRE detection **generates** an Error bit on the CEC line. CEC starts waiting for next valid Start-Bit at the end of the Error-Bit transmission. SBPE and LBPE errors are never set in case of BTE detection.
 - Note: can be set only if BRESTP=1
- **BRESTP: Stop on Bit Rising Error**
 - 0: BRE detection **only sets the BRE flag**. Rx-data bit is regularly sampled at 1.05ms and stored in the Rx-buffer.
 - 1: BRE detection **stops data reception**. CEC starts waiting for next valid Start-Bit immediately after BRE assertion or after Error-Bit generation if BREGEN=1. SBPE and LBPE errors are never set in case of BRE detection
 - Note: when this error is detected in a broadcast message, the behavior is more complex and controlled also by BRDNOGEN bit

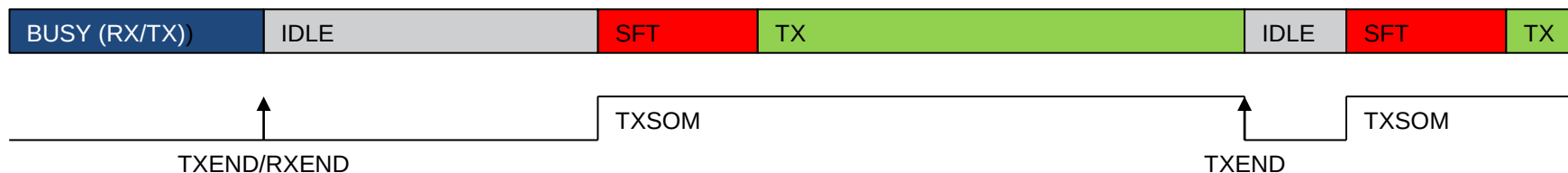
Bit Timing Error configured action 2/2

- **LBPEGEN: Generate Error-Bit on Long Bit Period Error**
 - 0: LBPE detection **does not generate** Error-Bit on the CEC line.
 - 1: LBPE detection **generates** an Error bit on the CEC line. CEC starts waiting for next valid Start-Bit at the end of the Error-Bit transmission.
 - Note: when this error is detected in a broadcast message, the behavior is more complex and controlled also by BRDNOGEN bit
- **BRDNOGEN: Not force Generate Error-Bit in Broadcast message**
 - 0: BRE or LBPE detection in a broadcast message **does generate** Error-Bit on the CEC line like if BREGEN or LBPEGEN are set.
 - 1: BRE or LBPE detection in a broadcast message **does not generate** Error-Bit on the CEC line if not configured by BREGEN and LBPEGEN bits.

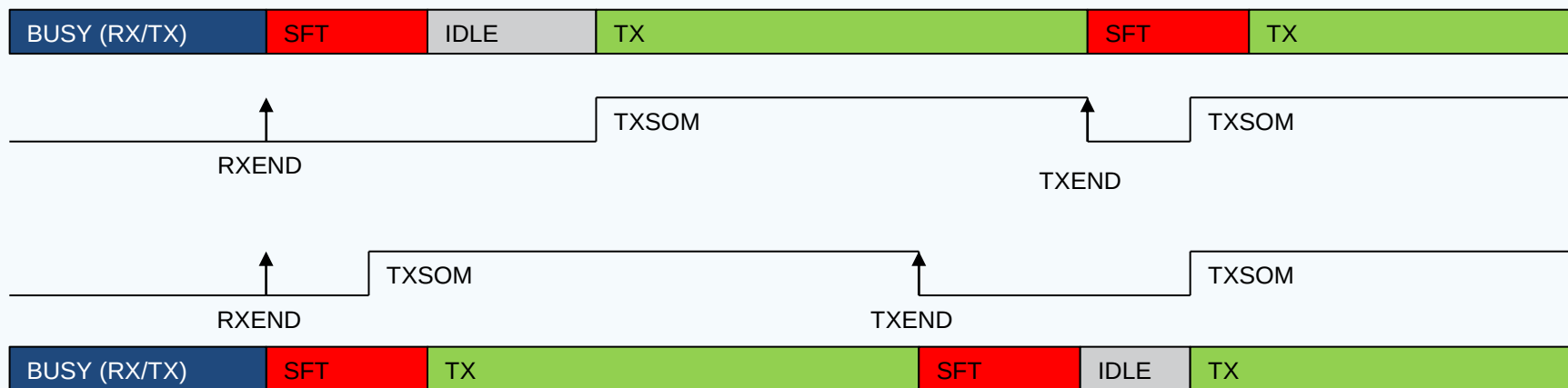
Signal Free Time (SFT) configuration

- Upon transmission command, CEC starts sending Start-Bit after the following number of nominal data bit periods of inactivity depends on SFT value:
 - 0x0: automatic HW control
 - 3: in case of previous transmission unsuccessful
 - 5: in case of new initiator
 - 7: in case of previous transmission successful
 - 0x1: 1
 - 0x2: 1.5
 - 0x3: 2
 - ...
 - 0xF: 7.5
- Note: SFT can be set only when TXSOM='0'

SFTOPT=0 counts SFT at TXSOM (TX Start Of Message command)



SFTOPT=1 counts SFT at TXEND/RXEND/TXERR/RXERR



STM32F37x CEC vs STM32F100 CEC

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Features	F100 CEC	F37x CEC
Supports HDMI-CEC v1.3a specification	x	x
APB clock -with PRESC frequency divider	x x	x no need
32 KHz CEC kernel with Dual clock - LSE - HSI/244		x x
TX missing acknowledge error	x	x
RX missing acknowledge error	x	x
Reception in Listen Mode		x
Rx Tolerance Margin - Standard - Extended	x	x x
Arbitration(Signal free Time) - Standard (by HW) - Aggressive (by SW)	x	x x
Arbitration Lost Detected flag/interrupt		x
Automatic transmission retry supported in case of arbitration lost		x
Multi-address configuration		x

- Which errors are handled by HDMI-CEC ?
- What are the possible clock sources in STM32F37x CEC ?
- What are the new features in STM32F37x CEC comparing to the CEC in STM32F100 devices ?



Thank you !

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